Component	Complexity	Measured cost (N: 50 → 1000)
KGC key initialization	$O(N^2)$	0.60 s → 241.85 s ($\times 403$, $\propto N^{2.02}$)
Storage overhead	$O(N)$	3.47 MB → 65.73 MB ($\times 19$, $\propto N^{0.98}$)